Source-Free Cross-Domain Scene Classification of Remote Sensing Images via Statistics Matching and Noise Adaptation

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***Abstract*— In recent years, in order to alleviate the performance degradation problem caused by domain drift in scene classification (SC) tasks, some unsupervised domain adaptation (UDA) methods have been introduced into the field of remote sensing images (RSIs). Such methods require simultaneous access to both source and target domain data during training. However, the large storage and transmission costs of RSIs limit the further application of these methods. To address this challenge, we investigate the task of source-free cross-domain SC (SFSC) for RSIs. In the model adaptation process, only the target domain dataset and the trained source domain model are used. Our approach consists of two parts: a distribution alignment strategy based on source domain model statistics matching and a noise adaptation strategy. In order to fully utilize the knowledge of the pretrained source domain model, we fix the classifier to get the feature distribution of the source domain, so that the target domain feature distribution is close to the feature distribution of the source domain. The noise adaptation layer is inserted after the classifier in order to improve the robustness of the model to noise-containing pseudo-labels, and the sample-wise noise transfer matrix is learned. Experimental results on 12 transfer tasks on the cross-scene dataset, and two transfer tasks on the cross-sensor dataset, to validate the effectiveness of our approach. Compared to traditional unsupervised domain adaptive methods, our method is able to achieve better performance under the condition of not accessing the source domain data.**

***Index Terms*— Domain adaptation, dynamic noise modeling, remote sensing image (RSI), scene classification (SC).**

I. Introduction

As HUMAN observation technology develops, the spatial resolution of remote sensing images (RSIs) is increasing and the information about features is becoming more and more abundant. Intelligent interpretation for RSIs has also evolved from the early pixel level or object level to the scene level [1]. The task of RSI scene classification (SC) aims to infer the covered scene information based on a given image, and this high-level semantic information (e.g., forests, industrial areas, and rivers) is more in line with human cognition [2]. Therefore, SC has become an active research direction and has been widely used in the fields of land use and urban planning [3].

In recent years, with the introduction of deep learning paradigms such as convolutional neural networks (CNNs), the performance of RSI SC has been significantly improved [4]. Deep learning-based SC methods, although free from the defects of hand-designed features are still some distance away from practical applications. On the one hand, it is a data-driven paradigm that relies on a large number of expensive and high-quality labeled samples. On the other hand, it requires the training data and test data to conform to the iid assumption, and when the distribution of test data and the distribution of training data are large, the performance of the SC model will be significantly degraded, a phenomenon known as domain shifting [5]. And the uniqueness of remote sensing data, i.e., different sensor parameters, acquisition scenes, acquisition time, and other factors [6], further expands the domain gap.

In order to improve the generalization ability and cross-domain classification performance of models, some feature-aligned [7], [8] or adversarial [9], [10] unsupervised domain adaptation (UDA) SC methods have been proposed. Feature alignment-based approaches narrow the domain gap through various distributional difference metrics, while adversarial training-based approaches achieve adaptation by confusing the features of the source and target domains so that the model learns domain-invariant feature representations. In order to achieve feature alignment or confrontation, as shown in Fig. 1(a), the framework of UDA specifies that it is necessary to be able to access the data of both source and target domains during model training. In practice, this condition is often difficult to fulfill, considering the privacy and security, storage cost, and transmission efficiency of the source domain data. It is impractical to save large-scale source domain data for training on some resource-constrained devices.

The contradiction between practical needs and UDA settings has inspired the new research direction of source-free domain adaptation (SFDA). The cost of transmitting pretrained source domain models is much lower compared to directly transmitting source domain data [11]. Fig. 1 illustrates the differences in SFDA and DA architectures. Due to the lack of source domain data, DA methods based on feature alignment or adversarial are difficult to use directly in the SFDA setup.

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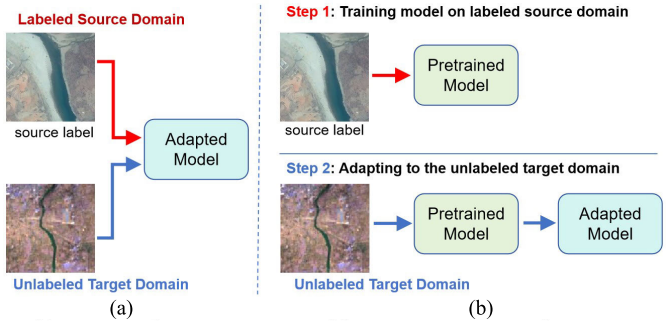


Fig. 1. Comparison of two domain adaptation frameworks, with unsupervised. (a) (Left) Domain adaptation and (b) (right) SFDA.

One intuitive approach is to synthesize the source domain data based on a priori knowledge and then use traditional DA methods [12]. Another approach is to use the target domain data based on the self-training framework [13], [14], [15] and adjust the source domain model directly. It is not difficult to find that the method based on source domain reconstruction needs to realize the synthesis of source domain data, which adds extra training time and storage cost. And the self-training-based methods, despite improving the quality of pseudo-labels through various methods, ignore the essential problem of the fitness of the target domain data to the pretrained source domain model, and can only achieve suboptimal results. On the other hand, existing SFDA methods are mainly designed for natural images, compared to RSIs where the cause of domain shift is more complex, containing unique influences such as external atmospheric radiation, spectral offsets, and internal sensor parameter differences, in addition to variations in target shapes and high-level textures, and thus more challenging.

In this work, aiming at solving the problem of unavailability of source domain data during the domain adaptation process of cross-domain SC, we introduce for the first time a RSI source-free cross-domain SC (SFSC) setup, where we utilize only unlabeled target domain data and pretrained source domain classification models to realize cross-domain SC. Specifically, we improve the self-training-based framework, called statistics matching and noise adaptation (SMNA) network. The framework consists of two parts: a distribution alignment strategy based on source domain model statistics matching and a noise adaptation module. On the one hand, it is believed that although the source domain data are not obtainable in SFSC, after pretraining, the knowledge related to the source domain data has been saved in the pretrained source domain model. In order to fully exploit the knowledge of the pretrained source domain model, during adaptation, we fix the classifier and train only the feature extractor. The feature distribution of the source domain is obtained from the batch normalization (BN) layer of the classifier, so that the target domain feature distribution is close to the feature distribution of the source domain, thus realizing the distribution alignment between the target domain data and the source domain data. On the other hand, in order to improve the robustness of the pseudo-label to noise, the noise adaptation module is inserted after the classifier, and a data-driven approach is used to adaptively learn the noise transfer matrix.

Our contribution in this article can be summarized as follows.

1) In order to compensate for the shortcomings of the existing cross-domain RSI SC settings, a more practical SFDA setting is proposed.
2) Estimate the source domain data distribution through the source domain model, so as to realize the distribution alignment between the target domain data and source domain data.
3) A dynamic noise adaptation module is introduced to capture different noise patterns in the pseudo-labels and improve the classification accuracy of the model for unlabeled target domain data.

The remainder of this article is organized as follows. Section II presents the literature work related to cross-domain SC and SFDA in computer vision. Section III details the overall process of SMNA and its components. Section IV presents experimental results on both cross-scene and cross-sensor datasets as well as extensive analytical results. Finally, we conclude this study in Section V.

II. Related Work

A. Cross-Domain SC

Cross-domain SC methods in remote sensing imagery, which continue the architecture of generalized domain adaptation, can be classified into two categories, i.e., based on distribution difference metrics and adversarial feature alignment. Distributional difference metrics aim to minimize the variance between the data distributions in the source and target domains. Zhu et al. [16] introduce the conditional distribution alignment strategy based on marginal distribution alignment, which improve the cross-domain classification accuracy of the model. Yang et al. [17] adaptively balance the weights of marginal distribution and conditional distribution for different samples in the process of domain alignment. Zhang et al. [18] introduce subspace correlation maximization, map the source domain data to the relevant subspace to retain the representation information of the source domain, and adopt dynamic statistical distribution registration strategy to learn an adaptive factor to balance the interclass and intraclass distribution differences between domains. Inspired by the idea of adversarial learning in GAN, some methods introduce domain discriminators to realize the learning of domain-invariant features. Zhang et al. [19] design a multilevel adversarial dropout module to enhance the generalization ability of the network. Zhu et al. [20] use adversarial training to learn discriminative features to separate public and private samples in the target domain, and match source and target samples for cross-scene adaptation. Zhu et al. [21] proposed that applying the difference of adversarial pair cosine to interdomain and intradomain distributions can alleviate the problem of negative transfer and implicitly align cross-scene distributions. The above UDA methods require access to both source and target domain data in order to perform distributional alignment during model training, so these methods cannot be directly used in a source-free domain setting.

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B. Source-Free Domain Adaptation

Considering data privacy and storage and transmission efficiency, researchers have generalized unsupervised domain adaptive methods to source domain-free scenarios. In order to fully utilize the source domain model, the existing methods can be classified into two main categories [11]: domain reconstruction-based methods and self-training-based methods. Domain reconstruction-based methods expect to reconstruct a new domain to supervise the target domain, which can be a virtual source domain, a perturbed target domain, and an uncoupled target domain. Source domain reconstruction-based methods focus on how to generate virtual source domain data to help model adaptation. 3c-GAN [12] is a collaborative class-conditional generation adversarial network for generating training data in the style of the target domain, containing discriminators that match the distribution of the target domain and class-conditional generators that incorporate semantic similarity constraints. A^{2}Net [22] is the first work to decouple the target domain and implement in-domain adversarial alignment. It adaptively divides the target domain samples into the source domain similar set and the source domain dissimilar set, and gradually eliminates the distributional differences between the two sets through adversarial learning. And the self-training-based approach uses pseudo-labeling to supervise the training of the target domain data and improve the performance of the model on the target domain. However, the pseudo-labels generated directly using the source domain model are noisy due to the presence of domain bias. Such methods, therefore, focus on how to generate high-quality pseudo-labels. SHOT [13] proposes a self-supervised pseudo-labeling strategy that uses weighted k-means clustering to generate feature prototypes from which pseudo-labels can be assigned to the target domain data. MBNS [23] proposes to approximate the distribution of unobserved source data using batch normalized statistics stored in a pretrained model, mutual information between features and classifier outputs is also maximized to further improve classification performance. Wang et al. [24] and Chang et al. [25] also adopted a similar idea, achieving domain adaptation by sharing the gamma/beta of the BN layer of the source domain and the target domain. G-SFDA [14] and NRC [15] introduce label consistency constraints between locally neighboring samples to further improve the pseudo-labels. DaC [26] divides the target domain samples into source-like samples and target-specific samples, and combines the advantages of these two major classes of methods by adopting global class clustering for the former's small pseudo-label noise, and a local structure learning strategy for the latter's large pseudo-label noise. CoWA [27] unites the predictive confidence of the model and the structural confidence of the data itself to construct a new sample confidence score that better distinguishes the importance of samples and improves the quality of pseudo-labels.

Distinguishing from the above methods, our approach focuses on how to better utilize the information in the source domain model and the estimation of the pseudo-label noise level.

III. Method

In this section, we first introduce the definition and notation of the SFSC problem, then the SMNA framework consisting of two parts, and finally the flow of the algorithm and the optimization objective.

A. Preliminary

We are given a source domain dataset $\mathcal{D}_s = \{(x_s^i, y_s^i)\}_{i=1}^{n_s}$, where $x_s^i \in \mathcal{X}_s$, $y_s^i \in \mathcal{Y}_s$, and y_s^i is the label corresponding to the sample x_s^i, which contains n_s labeled samples on K classes. We also have a target domain dataset $\mathcal{D}_t = \{(x_t^i, y_t^i)\}_{i=1}^{n_t}$, where $x_t^i \in \mathcal{X}_t$ which contains n_t unlabeled samples. $\mathcal{D}_s$ and $\mathcal{D}_t$ have the same label set.

The UDA task investigates how to make full use of the labeled source-domain data $\mathcal{D}_s$ to learn a model to predict the labels of the unlabeled target-domain data $\mathcal{D}_t$, using a joint source- and target-domain feature optimization framework.

While in this article, we are concerned with the SFDA problem of not obtaining $\mathcal{D}_s$ when adapting f_s to $\mathcal{D}_t$ using only the pretrained source domain model f_s without accessing the source domain data in the setting of the adaptation process, i.e. $f_t : \mathcal{X}_t \to \mathcal{Y}_t$, with only $\{(x_t^i)\}_{i=1}^{n_t}$ and the source function f_s.

The classification model f_s consists of a feature extractor $g_s : \mathcal{X}_s \to \mathbb{R}^d$ and a classifier $h_s : \mathbb{R}^d \to \mathbb{R}^K$ based on the fully connected (FC) layer, i.e., $f_s(x) = h_s(g_s(x))$. The target domain model can be represented accordingly as $f_t(x) = h_t(g_t(x))$.

B. Overview

As shown in Fig. 2, this article proposes a novel SFSC method. Due to the lack of source domain data, unlike traditional UDA algorithms, we cannot directly align the interdomain distribution. We propose the statistics matching and noisy adaptation strategy in order to solve this problem. First, an initial model is trained using labeled source domain data. Then, the pretrained source domain model weights are used to initialize the feature extractors and classifiers of the target domain model. For unlabeled target domain data, we improve the classification performance of the model in two ways. Considering that the target domain data has a different distribution from the source domain data, directly using these biased features may lead to the mismatch problem. To mitigate this problem, we propose the statistics matching strategy as shown by the red locked lock in Fig. 2, where we freeze the model's classifier, take the statistics of BN in the classifier as the distribution of the source domain dataset, and train only the model's feature extractor as shown by the red unlocked lock in Fig. 2, through the Wasserstein distance measures the distance between the target domain feature distribution and the source domain feature distribution to achieve distribution alignment. Second, given that there is a significant amount of noise in the pseudo-labels generated by the source domain model, using them directly to fine-tune the model will lead to suboptimal results. We propose the noise adaptation strategy, which inserts a dynamic noise adaptation module after the classifier so that each sample learns the noise transfer matrix

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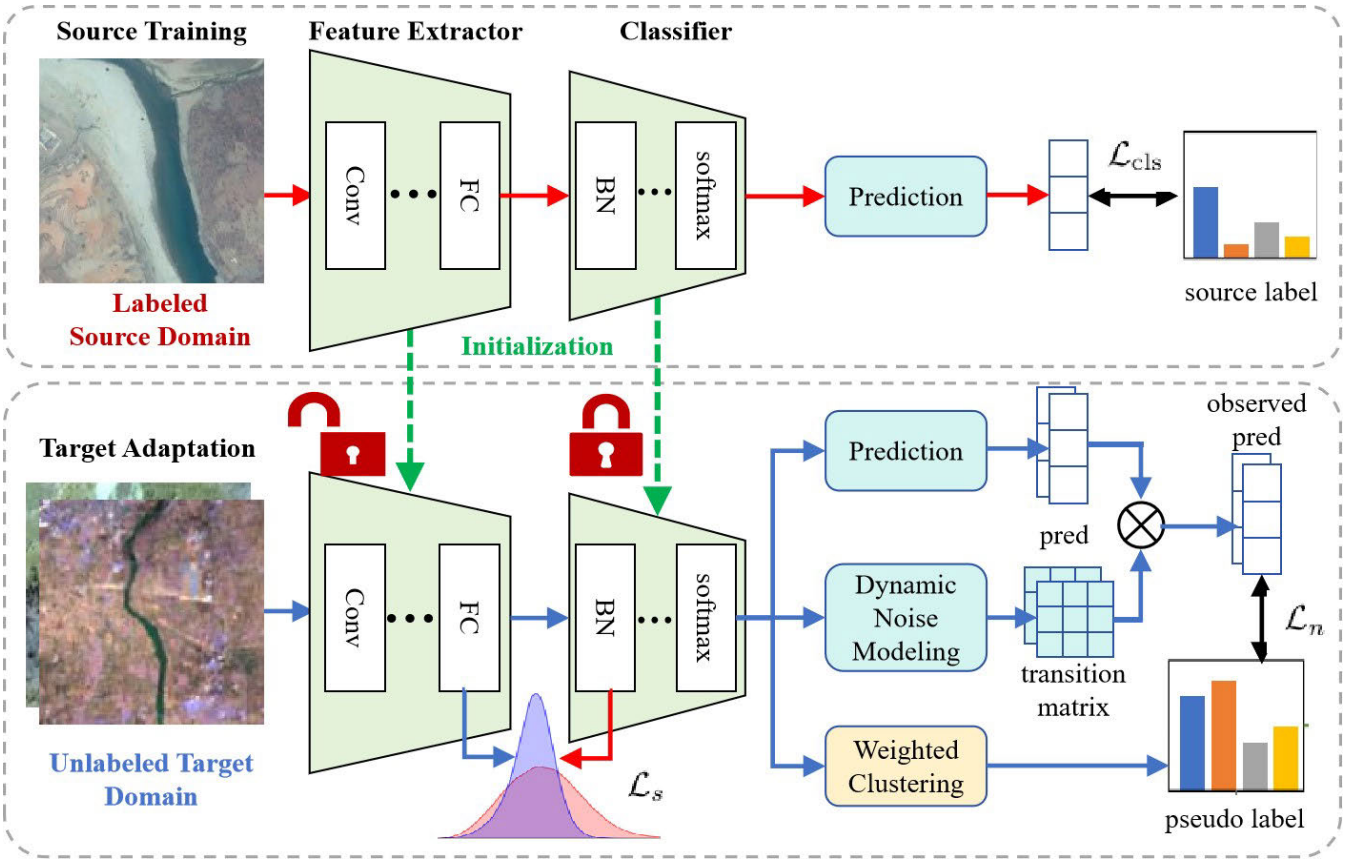


Fig. 2. Source domain model is trained using the labeled source domain data and the parameters of the target domain model are initialized. Freeze the classification layer of the target domain model and obtain approximate source domain feature distributions from it. For the feature extractor, impose a distribution alignment loss so that the feature distribution of the target domain approximates the feature distribution of the source domain. At the same time, a dynamic noise transfer matrix is introduced to capture the noise of different modes, which is fine-tuned using pseudo-labeling to improve the classification accuracy of the target domain model.

independently, expecting to cover the noise types of different patterns, improving the robustness of the model to the noise in the pseudo-labels, and ultimately improving the classification performance of unlabeled data.

C. Source Model Generation

We use the classical cross-entropy loss for optimization, only for labeled source domain data

$$\mathcal{L}_{\text{cls}} = -\frac{1}{n_s}\sum_{x_i\in\mathcal{D}_s}\sum_{k=1}^{K} y_{ik}\log\delta_k(h_s(g_s(x_i))) \tag{1}$$

where y_{ik} is the one-hot encoding of the label, h_s is the classifier, K is the number of categories in the dataset, and δ_k is the softmax activation function.

D. Distributional Alignment by Statistics Matching

1) Revisiting BN : BN layer [28] is a widely used technique to reduce internal covariate shifting, thus making deep neural network training faster and more stable. Given a small batch of data, the computation of the BN layer can be formalized as the following three steps:

$$\mu_{(c)} = \frac{1}{M}\sum_{m=1}^{M} F_m^{(c)} \tag{2}$$

$$\sigma_{(c)}^2 = \frac{1}{M}\sum_{m=1}^{M}\left(F_m^{(c)} - \mu_{(c)}\right)^2 \tag{3}$$

$$\hat{F}_m = \gamma\frac{F-\mu(c)}{\sqrt{\sigma_{(c)}^2+\epsilon}} + \beta. \tag{4}$$

Here, F_m is the input and $\hat{F}_m$ is the output of the BN layer. μ and σ are the mean and variance within a mini-batch. γ and β are two parameters learned during training.

In the training phase, the global mean and variance are calculated using the moving average method. The global mean and variance are calculated as follows:

$$\bar{\mu}^{t+1} = (1-\alpha)\bar{\mu}^t + \alpha\mu^t \tag{5}$$

$$\left(\bar{\sigma}^{t+1}\right)^2 = \left(1-\alpha\right)\left(\bar{\sigma}^t\right)^2 + \alpha\left(\sigma^t\right)^2. \tag{6}$$

In the testing phase, the estimated global mean and variance were used for normalization.

2) Distributional Alignment Loss: After reviewing the BN layer above, we can see that the trained BN layer stores the mean and variance of the entire dataset.[1] In view of this, we freeze the classifiers of the source-domain model and use the BN statistics stored in them to approximate the

[1]Implemented with running-mean and running-var in Pytorch.

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feature distribution of the source-domain data. Meanwhile, the feature distribution of the target domain data can be computed from the features output in the feature extractor. By measuring the difference between these statistics, the distributions of the source and target domain data can be approximated. Specifically, we use Gaussian distributions to model the feature distributions of the source and target domains, i.e., $S = \mathcal{N}(\mu_S, \sigma_S^2)$ and $T = \mathcal{N}(\mu_T, \sigma_T^2)$. For reasons of symmetry in distributional metrics, we use the two-Wasserstein distance [29] as a measure of the distance between the two distributions, rather than the Kullback–Leibler divergence. We can write the style loss when aligning the batch statistics of the target domain data with the stored BN statistics as

$$\mathcal{L}_s = D_{\text{wass}}(S, T) = \sqrt{\|\mu_S - \mu_T\|_2^2 + \|\sigma_S - \sigma_T\|_F^2}. \quad (7)$$

E. Pseudo-Label Generation

We use the weight clustering technique in SHOT [13] to generate pseudo-labels for the target domain samples. First, the prototype of each class in the target domain is obtained using weighted clustering

$$c_k = \frac{\sum_{x_t \in \mathcal{X}_t} \delta_k(f_t(x_t)) g_t(x_t)}{\sum_{x_t \in \mathcal{X}_t} \delta_k(f_t(x_t))}. \quad (8)$$

Then, the pseudo-labels $\hat{y}_t$ are obtained by the nearest neighbor centroid classifier

$$\hat{y}_t = \arg\min_k D_f(g_t(x_t), c_k). \quad (9)$$

Here, D_f is used to calculate the cosine distance.

Features extracted by backbone $g_t(x_t)$ are weighted averaged through the output probability $\delta_k(f_t(x_t))$ of the softmax layer of classifier to obtain the centroid of each category. Then, the centroid c_k with the closest distance is found for each sample and used as the pseudo-label $\hat{y}_t$ of the sample. Then, recalculate the centroid based on the updated sample pseudo-labels and update the pseudo-labels again. This process can be repeated multiple times.

F. Dynamic Noise Adaptation

1) Global Noise Modeling: The use of weighted clustering does not completely eliminate the noise interference in the pseudo-labeling. Since it describes the transformation from real to noisy labels, noise probabilistic modeling is a promising approach in robust training [30].

As shown in Fig. 3, in contrast to conventional classification modeling methods, noise probability modeling consists of a classifier, which predicts the category probability distribution $\hat{y}$, and a transfer matrix T, which is usually unknown and it needs to be estimated. The noise transfer matrix transforms the clean y into the observed category probability distribution $\hat{y}$, i.e., the label containing noise, which is formally defined as follows:

$$p(\tilde{y} \mid \boldsymbol{x}, \boldsymbol{w}) = \sum_{i=1}^{k} T_{ij}\, p(y = i \mid \boldsymbol{x}, \boldsymbol{w}) \quad (10)$$

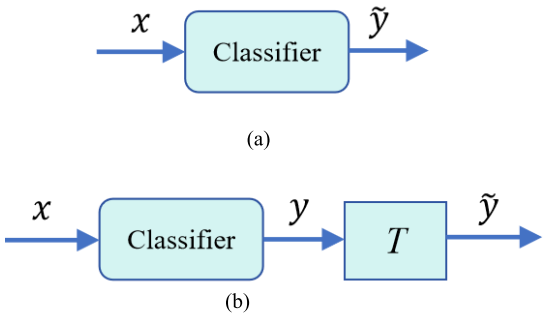


Fig. 3. Comparison of conventional classification modeling methods and noisy probabilistic modeling methods. (a) Ordinary modeling and (b) noise probabilistic modeling.

$$T_{ij} = p(\tilde{y} = j \mid y = i, \boldsymbol{x}) = p(\tilde{y} = j \mid y = i). \quad (11)$$

Therefore, the optimization objective of noise probability modeling can be defined as

$$\mathcal{L}_g = \frac{1}{N} \sum_{n=1}^{N} -\log \sum_{i}^{k} p(\tilde{y} \mid y = i)\, p(y = i \mid \boldsymbol{x}_n, \boldsymbol{w}). \quad (12)$$

2) Dynamic Noise Modeling: The classical noise probability modeling approach described above assumes that the noise labeling error depends on the clean labels, and is a global estimate that is overly simplistic. Considering that each sample possesses a different noise pattern, we dynamically generate a transfer matrix T for each sample, that characterizes the noise pattern of the sample, but with the parameter w shared across the entire dataset. Compared to a single model for global noise modeling, our approach adds only a few parameters and is able to adaptively characterize the noise pattern of each sample.

First, the features F of each sample are embedded using a FC layer to obtain Z, and then, we use softmax to compute the transfer matrix T for each sample

$$T_n = \frac{\exp(w_{ij}^{\mathrm{T}} \mathbf{z}_n + b)}{\sum_{j=1}^{K} \exp(w_{ij}^{\mathrm{T}} \mathbf{z}_n + b)} \quad (13)$$

where T_n is the conditional probability of an input sample $\boldsymbol{x}_n$ being labeled as category j, K is the number of categories, and w_{ij} is a weight vector characterizing the confusion between i and j.

We dynamically generate the w_{ij} corresponding to each sample based on the embedding of the sample

$$w_{ij} = \delta_k(v_{ij} * \mathbf{z}_n)$$

where v_{ij} is a learnable parameter. In practice, we randomly initialize v_{ij}. Because the noise transition matrix of each sample depends on the feature of each sample, this is what we call the dynamic generation process.

Similarly, multiplying the predicted probability with the noise transfer matrix yields the observed classification probability, which is used to match the noise labels after clustering, while the predicted probability serves as the output of our

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model at test time. Therefore, the final noise sample classification loss is

$$\mathcal{L}_n = -\frac{1}{n_t}\sum_{x_i\in\mathcal{D}_t}\sum_{k=1}^{K}\hat{y}_t \log T_n p(y=i \mid \boldsymbol{x}, \boldsymbol{w}). \tag{14}$$

Here, $\hat{y}_t$ is a pseudo-label containing noise generated by the pseudo-label generation module.

G. Overall Loss

Our source domain model consists of a feature extraction network and a classifier, and the target domain model consists of a feature extraction network, a classifier, and a noise adaptation module. The source domain model is optimized by ordinary cross-entropy loss. The parameters of the target domain model are initialized using the trained source domain model weights. The overall optimization objective of the target domain model consists of distributional alignment loss, noise classification loss, maximum entropy loss, and diversity loss. The formalization is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_n + (\mathcal{L}_{\text{ent}} + \mathcal{L}_{\text{div}}) + \lambda\mathcal{L}_s. \tag{15}$$

Here, the distribution alignment loss $\mathcal{L}_s$ is used to bring the distance between the target domain data distribution and the source domain data distribution closer, and the noisy label classification loss $\mathcal{L}_n$ is used to classify the pseudo-labeled data in the target domain containing noise. In addition, we follow the maximum entropy loss $\mathcal{L}_{\text{ent}}$ and $\mathcal{L}_{\text{div}}$ from SHOT [13] to give individual determinism and global diversity to the target output. The λ is a weight parameter to adjust the weight of the distributional alignment loss.

The detailed training procedure of the SMNA is shown in Algorithm 1.

Algorithm 1 Statistics Matching and Noise Adaptation Algorithm

Input: Source domain dataset $\mathcal{D}_s = \{(x_s^i, y_s^i)\}_{i=1}^{n_s}$ and target domain dataset $\mathcal{D}_t = \{(x_t^i, y_t^i)\}_{i=1}^{n_t}$;
Output: The learned weight parameters θ_t of the SMNA;

Stage 1 Source Model Generation
for $i = 1; i < n_s; i++$ **do**
  Calculate the source domain classification loss $\mathcal{L}_{cls}$ through Equation 1;
  Update $\theta_s \leftarrow \theta_s - l\frac{\partial \mathcal{L}_{cls}}{\partial \theta_s}$;
end for

Stage 2 Target Model Adaptation
Initialize the target domain model parameters θ_t using the trained source domain model weights θ_s.
for $j = 1; j < n_t; j++$ **do**
  Calculate the pseudo label $\hat{y}_t$ of target domain data through Equation 10;
  Calculate the noisy classification loss $\mathcal{L}_n$ through Equation 15;
  Calculate the distribution alignment loss $\mathcal{L}_s$ between F^s and F^t through Equation 8;
  Calculate the maximum entropy loss $\mathcal{L}_{ent}$ and $\mathcal{L}_{div}$ from SHOT [13];
  $\mathcal{L}_{total} = \mathcal{L}_n + (\mathcal{L}_{ent} + \mathcal{L}_{div}) + \lambda\mathcal{L}_s$;
  Update $\theta_t \leftarrow \theta_t - l\frac{\partial \mathcal{L}_{total}}{\partial \theta_t}$;
end for

IV. Experiments

A. Description of Datasets

Due to the lack of available remote sensing cross-domain SC datasets, in order to verify the effectiveness of the proposed method, we construct multiple cross-domain datasets, including one cross-scene dataset and two cross-sensor datasets.

For the cross-scene dataset [31], four SC datasets collected from Google Earth, namely AID [32], CLRS [33], MLRSN [34], and RSSCN7 [35], are combined. Specifically, the AID dataset covers 30 typical scene categories with 220–420 images per category, each processed at 600 × 600 and with a pixel resolution of 0.5–8m. The CLRS dataset covers 25 typical scene categories, each with 600 images, each processed at 256 × 256, with a pixel resolution of 0.26–8.85 m. The MLRSN dataset covers 46 typical scene categories with 1500–3000 images per category, each processed at 256 × 256 and with a pixel resolution of 0.1–10m. The RSSCN7 dataset covers seven typical scene categories; each category has 400 RSIs; each image is processed at 400 × 400; pixel resolution is 0.5–8 m, divided into four different scales, and each scale has 100 images.

Based on the description of the above four remote sensing datasets, seven public categories of farmland, forest, industry, grassland, parking lot, housing, and river are reintegrated. Therefore in Table I, the cross-scenario dataset contains 2500 AID (a) samples, 4200 CLRS (C) samples, 14 897 MLRSN (M) samples, and 2800 RSSCN7(R) samples.

TABLE I
Detailed Introduction to the Cross-SC Dataset

Classes	AID	CLRS	MLRSN	RSSCN7
Farmland	370	600	2501	400
Forest	250	600	2501	400
Industrial	390	600	2110	400
Meadow	280	600	2502	400
Parking	390	600	2502	400
Residential	410	600	2783	400
River	410	600	2500	400
Resolution(m)	0.5-8	0.26-8.85	0.1-10	0.5-8
Size	600 × 600	256 × 256	256 × 256	400 × 400
Main Source	Google Earth	Google Earth	Google Earth	Google Earth

For cross-sensor dataset I, two datasets from NWPU-RESISC45 [2] and NaSC-TG2 [36] are combined. Specifically, the NWPU-RESISC45 dataset covers RSIs from 45 typical scene categories collected from Google Earth, with 700 images per category, each processed at 256 × 256 and with a pixel resolution of 0.2–30m. The NaSC-TG2 dataset covers ten typical scene categories such as beaches, circular farmlands, clouds, deserts, forests, mountains, rectangular farmlands, residential areas, rivers, and snow-capped mountains, and unlike other SC datasets, this dataset was collected by the imaging spectrometer in China's Tiangong-2 space laboratory. Each image is processed at 128 × 128 and has a pixel resolution

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TABLE II
DETAILED INTRODUCTION TO THE CROSS-SENSOR CLASSIFICATION DATASET I

Classes	NWPU-RESISC45	NaSC-TG2
Beach	700	2000
CircularFarmland	700	2000
Cloud	700	2000
Desert	700	2000
Forest	700	2000
Mountain	700	2000
Rectangular farmland	700	2000
Residential	700	2000
River	700	2000
Snowberg	700	2000
Resolution(m)	0.2-30	100
Size	256×256	128×128
Source	Google Earth	TianGong-2

TABLE III
DETAILED INTRODUCTION TO THE CROSS-SENSOR CLASSIFICATION DATASET II

Classes	WHU-RS19	EuroSAT
Farmland(AnnualCrop)	50	3000
Forest	50	3000
Industry	50	2500
Meadow(Pasture)	50	2000
Residential	50	3000
River	50	2500
Resolution(m)	0.5	10-60
Size	600×600	64×64
Source	Google Earth	Sentinel-2

of 100 m. The samples of the cross-sensor dataset are shown in Fig. 4.

Based on the above description of the two remote sensing datasets, ten categories shared by the two categories in NWPU-RESISC45 are selected as experimental data based on the ten categories in NaSC-TG2 dataset. Therefore as shown in Table II, the composite cross-sensor classification dataset contains 27 000 images, among which 7000 are NWPU-RESISC45 (N) and 20 000 are NaSC-TG2 (T). Compared with the composite cross-SC dataset, the images in this dataset come from two sensors, with large differences in imaging parameters and spectral range, and the huge difference in spatial resolution (30 and 100 m) poses a greater challenge to the cross-domain SC algorithm.

For cross-sensor dataset II, as shown in Table III, two datasets from WHU-RS19 [37] (W) and EuroSAT [38] (E) are combined. Specifically, the WHU-RS19 dataset contains RSIs of 19 land-use types extracted from Google Earth and released by Wuhan University in 2012, with an image size of 600×600. Based on Sentinel-2 satellite images, the EuroSAT dataset covers 13 spectral bands and consists of ten categories, each of which contains 2000–3000 RSIs, each with a size of 64 $\times$ 64 pixels. Based on the above description of the two remote sensing datasets, six categories of farmland, forest, industry, grassland, housing, and river shared by the two are taken as experimental data.

B. Implementation Details

In order to fairly compare the performance of different algorithms, we adopt the same network architecture and training strategy as the baseline algorithm SHOT [13] for all algorithms. Specifically, the pretrained ResNet-50 [39] on ImageNet [40] is used as the backbone network for the cross-scene and cross-sensor benchmarks.

To achieve alignment between the source and target domains during target model adaptation, we freeze the weights of the target classifier $h_t = h_s$ and only update the target feature extractor g_t that is initialized from g_s. The batch size is set to 64 for both benchmark datasets. We set the learning rate to 0.01 for cross-scene and cross-sensor, and train 15 epochs for two datasets. On the cross-scene and cross-sensor datasets, we uniformly set the weight λ of the distribution alignment loss to 0.1. We use the open-source PyTorch-1.8.1 [41] framework for the implementation of all methods, and train and test on an NVIDIA GeForce RTX 2080Ti GPU equipped with 11GB memory.

We use the classic overall accuracy (OA) as an evaluation metric for UDA and SFSC. It is expressed as follows:

$$\mathrm{OA} = \frac{|x : x \in \mathcal{D}_t \wedge F(x) = y|}{|x : x \in \mathcal{D}_t|}. \tag{16}$$

TABLE IV
ACCURACY (%) OF THE DIFFERENT METHODS ON THE TWO TRANSFER TASKS IN THE CROSS-SENSOR CLASSIFICATION DATASET, WITH THE AVERAGE ACCURACY IN THE LATER COLUMN

Method	Venue	SF	N→T	T→N	AVG
ERM	-	✘	61.25	63.21	62.23
DAN	PMLR15	✘	73.59	77.66	75.62
DAAN	ICDM19	✘	62.81	64.19	63.50
DANN	PMLR15	✘	77.93	92.24	85.09
DeepCoral	ECCV16	✘	66.72	65.66	66.19
DSAN	TNNLS20	✘	95.53	93.54	94.54
BNM	CVPR20	✘	81.74	68.36	75.05
AMRAN	TGRS21	✘	64.28	90.10	77.19
MRDAN	TGRS22	✘	85.60	91.13	88.37
APA	AAAI23	✘	68.48	57.04	62.76
SHOT	ICML20	✔	89.80	91.41	90.61
MBNS	Arxiv21	✔	91.99	92.66	92.33
NRC	NeurIPS21	✔	89.73	85.91	87.82
G-SFDA	ICCV21	✔	67.42	66.34	66.88
CoWA	ICML22	✔	88.16	94.56	91.36
DaC	NeurIPS22	✔	83.58	83.41	83.50
AaD	NeurIPS22	✔	89.84	84.01	86.93
SMNA	ours	✔	**99.01**	**94.66**	**96.84**

TABLE V
ACCURACY (%) OF THE DIFFERENT METHODS ON THE TWO TRANSFER TASKS IN THE CROSS-SENSOR CLASSIFICATION DATASET II, WITH THE AVERAGE ACCURACY IN THE LATER COLUMN

Method	Venue	SF	E→W	W→E	AVG
ERM	-	✘	73.40	37.29	55.35
DAN	PMLR15	✘	78.59	72.97	75.78
DAAN	ICDM19	✘	76.45	45.05	60.75
DANN	PMLR15	✘	76.76	82.29	79.53
DeepCoral	ECCV16	✘	74.01	52.67	63.34
DSAN	TNNLS20	✘	87.46	84.74	86.10
BNM	CVPR20	✘	77.68	83.12	80.40
SHOT	ICML20	✔	88.05	41.24	64.65
NRC	NeurIPS21	✔	84.42	94.78	89.60
G-SFDA	ICCV21	✔	77.72	50.18	63.95
CoWA	ICML22	✔	81.63	57.05	69.34
AaD	NeurIPS22	✔	85.81	**95.58**	90.70
SMNA	ours	✔	**99.19**	95.19	**97.19**

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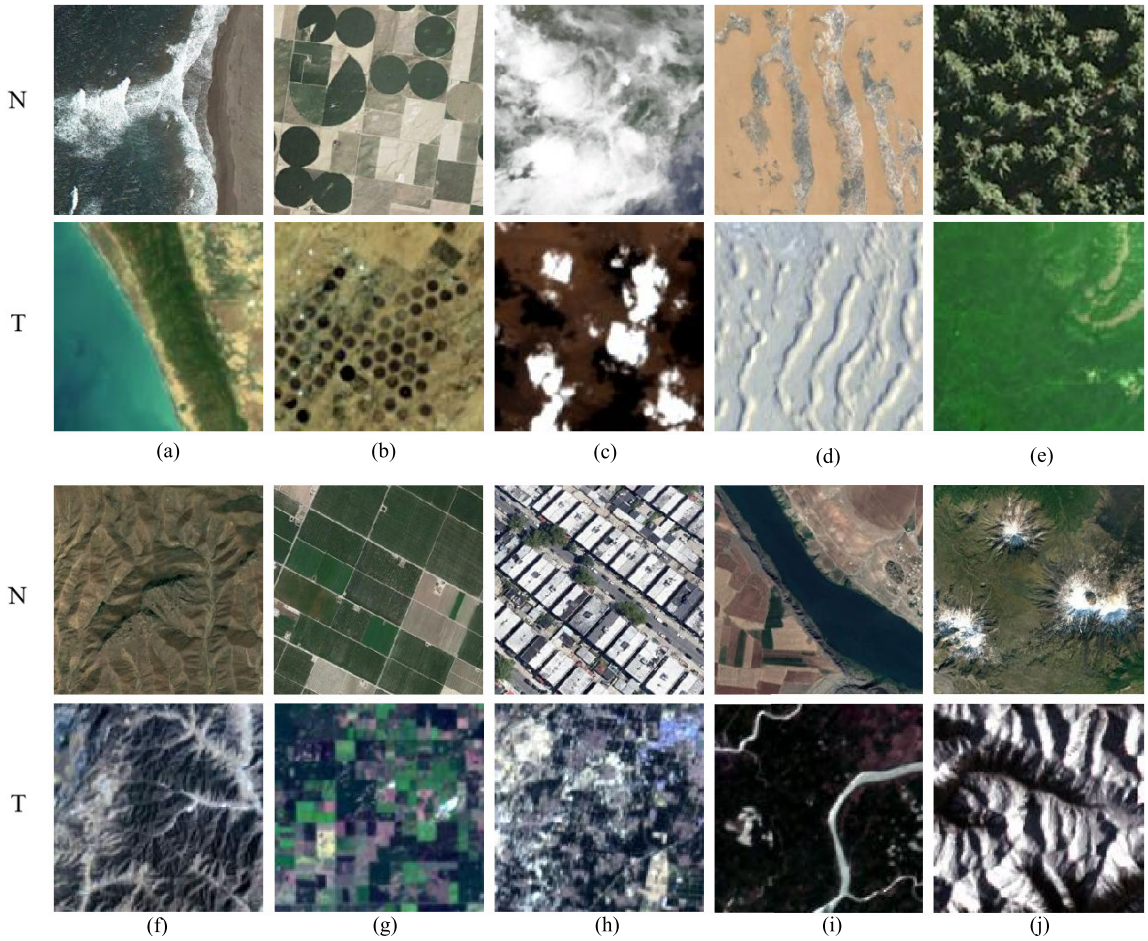


Fig. 4. We use the NWPU-RESISC45 dataset (first row) and the NaSC-TG2 dataset (second row) to form the cross-sensor dataset. The corresponding categories for both datasets are shown in each column, i.e., (a) beach, (b) circular farmland, (c) cloud, (d) Desert, (e) forest, (f) mountain, (g) rectangular farmland, (h) residential, (i) river, and (j) Snowberg.

TABLE VI

ACCURACY (%) OF THE DIFFERENT METHODS ON THE 12 TRANSFER TASKS IN THE CROSS-SC DATASET, WITH THE AVERAGE ACCURACY IN THE LATER COLUMN

Method	Venue	SF	A→C	A→M	A→R	C→A	C→M	C→R	M→A	M→C	M→R	R→A	R→C	R→M	AVG
ERM	-	✘	90.31	73.73	69.79	96.44	75.14	69.64	72.80	65.62	62.32	81.28	78.31	62.30	74.81
DAN	PMLR15	✘	92.17	83.73	73.61	96.24	80.68	73.32	75.92	76.00	70.79	91.12	84.90	73.84	81.03
DAAN	ICDM19	✘	91.05	76.87	70.29	96.40	76.17	68.93	74.52	65.79	62.07	83.08	78.90	64.33	75.70
DANN	PMLR15	✘	91.38	83.40	75.82	97.00	82.85	80.21	71.12	80.12	80.50	94.36	90.38	74.33	83.46
DeepCoral	ECCV16	✘	90.79	76.78	71.18	96.32	77.83	70.25	74.48	69.21	62.82	83.56	79.02	66.38	76.55
DSAN	TNNLS20	✘	95.19	83.81	80.68	98.72	85.18	80.00	75.48	79.24	78.07	95.64	90.19	85.84	85.67
BNM	CVPR20	✘	93.67	83.61	77.11	98.64	79.11	78.61	70.80	74.95	72.07	96.00	88.05	84.08	83.06
AMRAN	TGRS21	✘	91.29	81.22	75.43	96.52	81.82	76.64	71.09	74.48	74.71	90.28	85.17	81.40	81.67
MRDAN	TGRS22	✘	94.14	82.63	80.32	97.44	81.23	81.68	76.28	76.36	79.43	93.32	87.50	73.00	83.61
APA	AAAI23	✘	93.52	77.14	75.64	98.44	76.00	74.86	81.32	61.45	60.71	97.68	86.88	67.79	79.29
SHOT	ICML20	✔	95.05	85.82	**81.93**	**98.94**	84.60	79.86	70.73	76.24	82.50	98.47	92.64	75.02	85.15
MBNS	Arxiv21	✔	95.10	85.59	81.21	40.92	84.70	81.68	70.00	76.00	83.25	98.77	92.55	74.55	80.36
NRC	NeurIPS21	✔	93.55	86.83	79.11	98.38	83.70	79.36	73.89	75.00	73.61	97.93	90.81	82.61	84.57
G-SFDA	ICCV21	✔	90.83	77.49	72.18	97.96	80.44	79.57	**83.93**	71.60	67.39	97.57	89.33	72.71	81.75
CoWA	ICML22	✔	91.02	82.57	76.46	97.60	81.24	77.57	83.80	**83.79**	67.82	93.20	76.02	77.27	82.36
DaC	NeurIPS22	✔	95.88	85.40	80.71	97.96	84.34	81.61	75.43	77.36	**84.61**	97.36	**94.14**	76.50	85.94
AaD	NeurIPS22	✔	94.19	**87.93**	78.54	98.78	82.67	78.71	71.35	75.07	74.21	98.37	92.14	90.31	85.19
SMNA	ours	✔	**95.95**	87.26	80.75	98.69	**91.56**	**82.61**	70.74	78.17	80.43	**98.98**	93.93	**90.92**	**87.50**

C. Baselines

Our comparison algorithms include representative UDA algorithms, as well as advanced SFDA algorithms. The UDA algorithms include ERM, DAN [42], DANN [43], DeepCoral [8], BNM [44], DSAN [45], AMRAN [16], MRDAN [46], APA [47]. SFDA algorithms include: SHOT [13], MBNS [23], NRC [15], G-SFDA [14], DaC [26], AaD [48], and CoWA [27]. It should be pointed out that the results we reported are the results of rerunning on the two datasets of cross-scene and cross-sensor. Since most of the comparison

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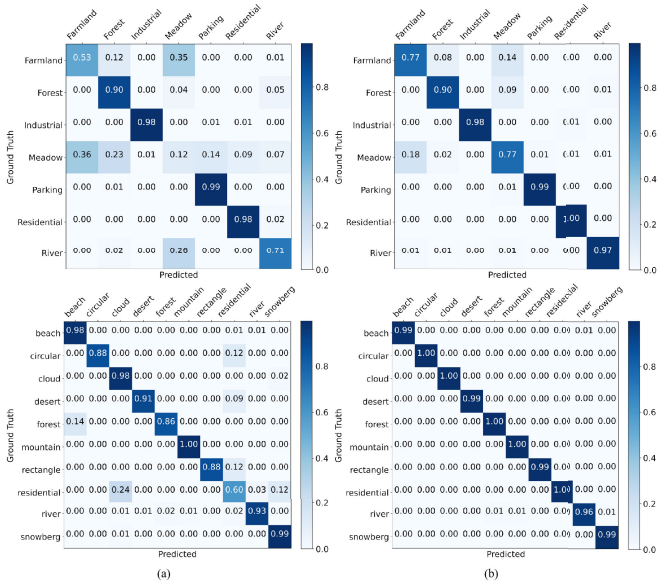


Fig. 5. Comparison of the confusion matrix between the SHOT method and the proposed SMNA method, with R→M transfer tasks in the first row and N→T transfer tasks in the second row. (a) SHOT and (b) SMNA.

TABLE VII
ABLATION EXPERIMENT RESULTS ON EACH MODULE OF SMNA

Method	PL	SM	NA	cross-scene	cross-sensor I
Source-only				74.81	62.23
SMNA	✔			85.15	90.61
	✔	✔		85.60	91.02
	✔		✔	87.01	96.79
	✔	✔	✔	87.50	97.19

TABLE VIII
COMPARISON OF COMPLEXITY, AND TRAINING TIME OF DIFFERENT MODELS ON THE EUROSAT TO WHU-RS19 TASK

Method	Acc (%)	GPU. (MiB)	Training (s)
SHOT	88.05	7585	120.44
NRC	84.42	9445	87.31
G-SFDA	77.72	9427	88.04
CoWA	81.63	7697	87.84
AaD	85.81	9447	87.4
SMNA	89.41	7595	119.4

algorithms are in the field of computer vision, we use the corresponding open-source code.

D. Comparison With State-of-the-Art Methods

Table IV shows the results of our approach and advanced algorithms on cross-sensor datasets. Our proposed SMNA strategy achieves the best average result of 96.84% on the two transfer tasks, as well as the best result on a single task. In particular, on the N→T task, our results are substantially ahead of other methods, improving by 4.5% over DSAN, the second best result, which is also based on a pseudo-labeling approach, illustrating the effectiveness of our dynamic noise modeling scheme for pseudo-labeling.

Table V shows the experimental results of different methods on cross-sensor dataset II. In terms of both transfer tasks and average accuracy, our proposed method achieves the best results, significantly outperforming traditional unsupervised cross-domain classification methods.

The results of our approach and the state-of-the-art algorithms on the cross-SC dataset are shown in Table VI. The upper half of the table shows the results of the traditional UDA approach and the lower half of the table shows the results of the SFDA approach, with the difference being that the former requires access to the source domain data during the adaptation process, while the latter does not. It is denoted by bolding the best results and underlining the second best results. Our proposed SMNA strategy has the highest average accuracy of 87.50% on the 12 transfer tasks, which is 1.56% higher than the next best DaC method. Our method achieves optimal or suboptimal results on nine tasks with AID, CLRS, and RSSCN7 as source domains. In addition, we found that the best results on individual transfer tasks were all produced by SFDA's method rather than the traditional UDA method, which, we believe, may suffer from overfitting to the source domain data during model adaptation. As for the average accuracy, the numerous algorithms centered on the pseudo-labeling technique are substantially ahead of the simple regularization methods, indicating that the pseudo-labeling technique is still of great potential when dealing with the domain adaptation problem. All of these results show that our approach achieves better results in more demanding situations, which is very convenient for practical applications where source domain data are difficult to transmit and access.

For the constructed cross-scene datasets, the resolutions of the AID, CLRS, and MLRSN datasets are approximately the same. The main factor affecting the cross-domain generalization of the model is the feature difference in different scenarios. Although the resolution of RSSCN7 is also 0.5–8m, it contains four different scales. When R is used as the source domain data, our method achieved optimal or suboptimal results in all three migration tasks and significantly outperformed other algorithms, proving that our method is insensitive to resolution changes. The same conclusion can also be drawn from the experimental results of cross-sensor datasets. The resolution of the NWPU dataset is 0.2–30m, while that of the TG2 dataset is 100 m. Both of these datasets are visible light images. However, the huge resolution difference brings challenges to the transfer of scene recognition capabilities. Our method still significantly outperforms other algorithms.

Fig. 5 shows the confusion matrices of SHOT [13] and our method on the R→M task in the first row, and the confusion matrices of SHOT and our method on the N→T task in the second row. It can be seen that the confusion matrix of our method is more centralized diagonally than the SHOT method. In particular, for the two visually very similar categories "farmland" and "meadow," SHOT easily recognizes "meadow" as "farmland," resulting in misclassification, and for "residential" and "cloud," two categories that often co-appear together, SHOT is also prone to confusion. Our adaptation strategy can effectively reduce the above two situations and enable more accurate cross-SC.

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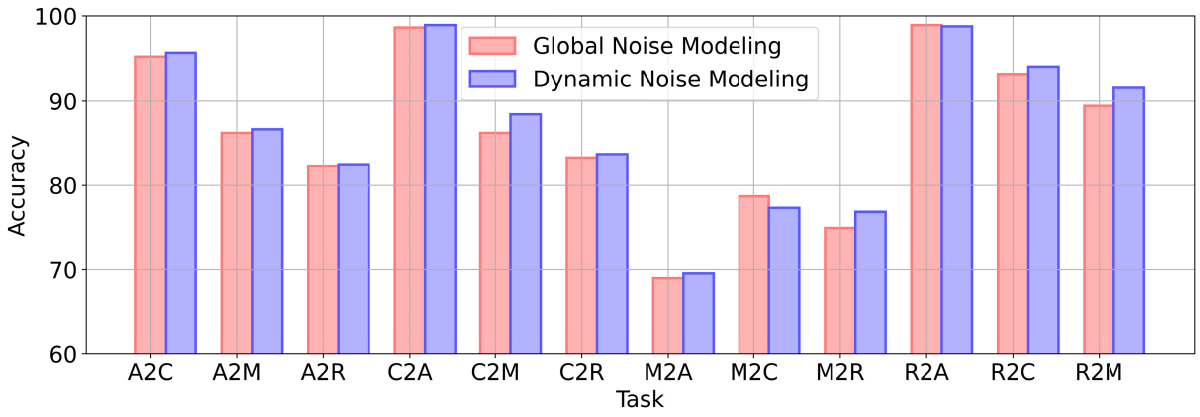


Fig. 6. Comparison of results between global noise modeling methods and dynamic noise modeling methods.

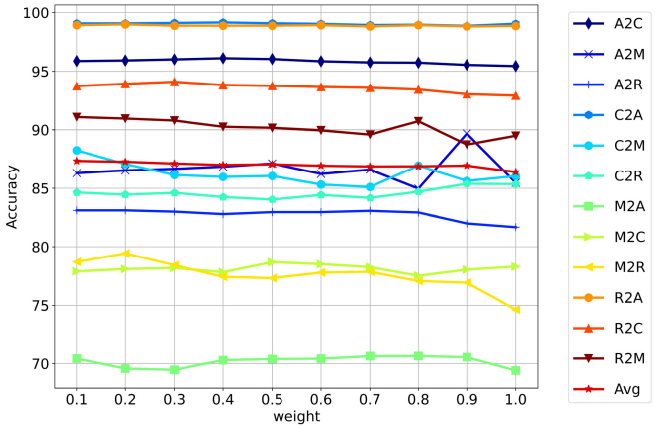


Fig. 7. Comparison plot of results when the model uses distribution alignment loss with different weights under the cross-scene transfer task.

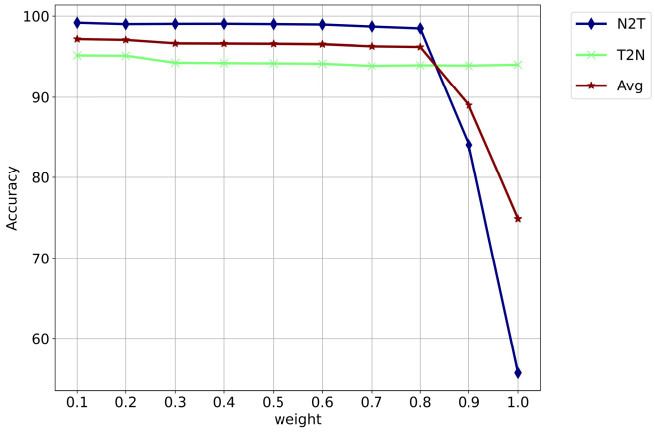


Fig. 8. Comparison of distribution matching losses with different weights on cross-sensor dataset.

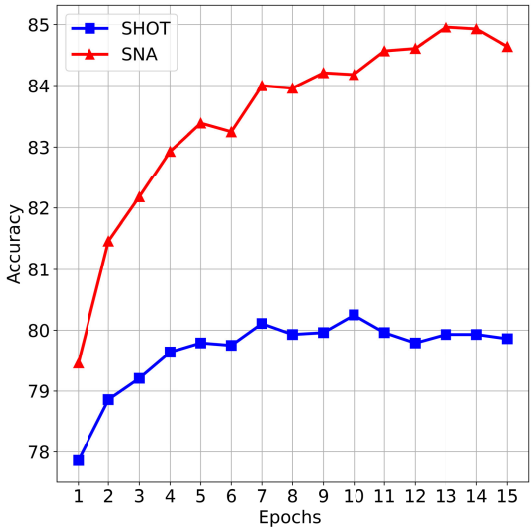


Fig. 9. Comparison plots of model accuracy in different epochs between SHOT and SMNA methods under the C→R transfer task.

E. Ablation Study

1) Effectiveness of each module: In Table VII, we report the results of the ablation study of our method on the cross-scene and cross-sensor datasets for each component. Where PL stands for pseudo labeling, SM stands for statistics matching, and NA stands for noise adaptation. Only the source domain model is used, while the results of direct testing on the target domain data are 74.81% and 62.23%, respectively. The baseline is the SHOT with pseudo-labeling fine-tuning used, and the results are 85.15%, and 90.61%, respectively. It can be found that the pseudo-label strategy can generate high-quality pseudo-labels because it uses pseudo-label clustering operations, and thus can significantly improve the classification performance of images in the target domain. Based on this, by adding SM, we improve the accuracy of cross-scene and cross-sensor by 0.45% and 0.41%, respectively. The statistical matching strategy is used to initialize and guide the feature learning process of the target domain dataset with the statistical information of the source domain dataset, reducing the domain gap between the source domain dataset and the target domain dataset. By adding NA, we improve the accuracy of cross-scene and cross-sensor by 1.86% and 6.18%, respectively. The use of the noise adaptation module can significantly alleviate the interference of noisy pseudo-labels on the data learning process of the target domain. These improvements validate the effectiveness of the distribution alignment strategy based on statistic matching and noise adaptation strategy of our proposed method. It also reveals

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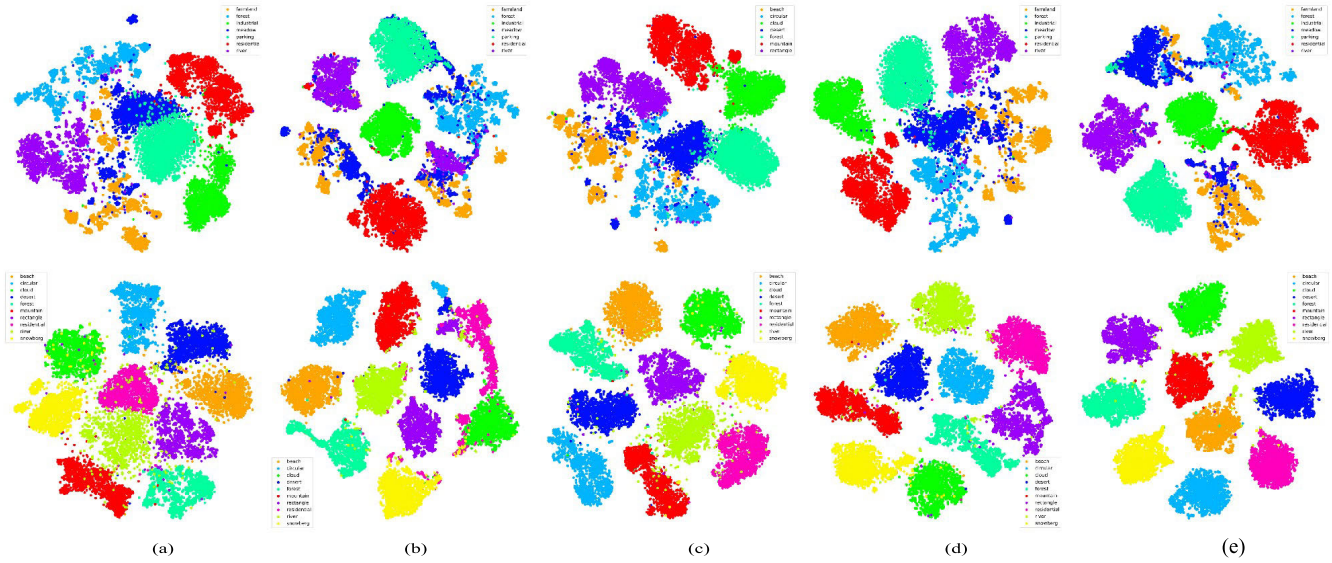


Fig. 10. Visualization results of t-SNE on the transfer task, (a) source-only, (b) SHOT, (c) G-SFDA, (d) NRC, and (e) ours method on the target domain samples. The first row is R→M and the second row is N→T.

that the two adaptation strategies are complementary and together they significantly improve the performance, by applying both strategies simultaneously we improve the accuracy of cross-scene and cross-scene by 2.19% and 6.58%, respectively. This reveals that the two adaptation strategies are complementary and that they work together to significantly improve performance.

2) Different noise modeling approaches: As shown in Fig. 6, we compare the results of traditional methods based on global noise modeling and dynamic noise modeling. The difference between the two modeling approaches is mainly in the establishment of the noise transfer matrix, where the global noise modeling approach learns a global noise transfer matrix, whereas the dynamic noise modeling strategy enables each sample to independently learn the noise transfer matrix. It can be seen that our proposed dynamic noise modeling strategy yields better results on the vast majority of the transfer tasks, which can cover different modes of noise types, improve the robustness of the model to the noise in the pseudo-labels, and ultimately improve the classification performance of unlabeled data.

3) Parameter sensitivity analysis: In our approach, λ is the weight that controls the loss of distributional alignment. In this experiment, we utilize different values of λ to analyze the sensitivity of the hyperparameters, as shown in Fig. 7, with experiments conducted at intervals of 0.1 from 0 to 1. It is observable that the accuracy of our method is insensitive to the value of λ for most transfer tasks. Only for the A→M task, there is a large jump in the results when $\lambda = 0.9$. Looking at the average results, if λ is set to 0.1, our method can produce better results. The parameter analysis results on the cross-sensor dataset, as shown in Fig. 8, also indicate that taking lambda as 0.1 can achieve the best model performance.

4) Robustness comparisons with SHOT: As shown in Fig. 9, in the process of model training, our method is always ahead of SHOT, which may overfit the noise in the pseudo-labels, causing the model to fall into a local optimal solution, which leads to a low generalization ability. By contrast, it is believed that our method is more robust and can overcome the pseudo-label noise problem and gradually improve the classification accuracy of the model.

5) Visualization of feature distribution: We use the widely used t-SNE method [49] to visualize features from different models as a way to gain insight into the effectiveness of our method. Specifically, the features output from the bottleneck layer of the model are captured for embedding. Fig. 10 illustrates the t-SNE results of different algorithms on the target domain samples. From left to right, they are source-only, SHOT, G-SFDA, NRC, and our method. The first row shows the results of the R→M transfer task in the cross-scene dataset, and the second row shows the results of the N→T transfer task in the cross-sensor dataset. Different colors are used in the figure to represent different classes of sample features. Overall, our method is able to achieve tighter clustering of features of the same class, as well as dispersion of features of different classes, and can significantly reduce the generation of outliers.

6) Comparison of model efficiency: Table VIII reports the model complexity and training of our proposed SMNA, which we use the EuroSAT to WHU-RS19 task as the basis for our evaluation. It can be observed that the existing SFSC models are roughly divided into two categories. One is the self-training method based on pseudo-label generation. The model complexity of this method is low, similar to that of ResNet-50 network, and the training time is longer because of the operation of pseudo-label generation. Another kind of method is the feature similarity measure as a constraint, which has a high memory occupation because the historical features need to be stored during the training process. The method proposed in this article belongs to the first class, and because of the existence of noise adaptation module, the video

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memory occupation is slightly increased compared with the SHOT method.

7) Limitation: Aiming at the task of SFSC, this article introduces a noise-based modeling method based on the self-training framework to improve the robustness of the model to noisy pseudo-labels. This method still has some limitations: on the one hand, the noise transfer matrix of the noise adaptation module is randomly initialized, and the whole learning process lacks constraints. Refer to article [50], and rank constraints can be introduced to restrict the noise matrix. On the other hand, for remote sensing data, the pseudo-label generation method based on weighted clustering can generate better pseudo-labels at the initial stage, which is conducive to the learning of noise adaptation module, while for visual recognition data, it cannot generate better pseudo-labels at the initial stage, which limits the learning of noise adaptation module. Data neighborhood structure information [15] can be introduced to improve the quality of the initial pseudo-label.

V. Conclusion

In view of the problem that the existing cross-domain SC methods need to access both source domain and target domain data during training, and the storage and transmission cost of remote sensing data is large, the task of SFSC of RSIs is studied. In the process of model adaptation, only the target domain data and the trained source domain model are used. In order to make full use of the knowledge of the pretrained source domain model, the classifier is modified, the feature distribution of the source domain is modeled, and the feature distribution of the target domain is close to that of the source domain by statistical matching method. In addition, a noise adaptive layer is inserted behind the classifier to improve the robustness of the model to noisy pseudo-labels by learning the sample noise transfer matrix. Experiments on cross-scene and cross-sensor datasets prove that the proposed method can achieve better performance without accessing the source domain data than the traditional UDA method. In the future work, better pseudo-label generation strategies will be studied to further improve the accuracy of SFSC methods.

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